

Regioselective Synthesis of Trialkylpyrazines via Nickel-Catalyzed Negishi Cross-Coupling of Pyrazine Triflate

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Abstract: A regioselective synthesis of trialkylpyrazines via nickel-catalyzed cross-coupling reaction of pyrazine triflate is reported. The 5-substituted 2,3-dimethylpyrazine derivatives including trail pheromone components of the ant *Eutetrarmorium mocquersyi* have been successfully synthesized in good yields by nickel-catalyzed Negishi cross-coupling reactions of pyrazine triflate mediated by alkyl and arylzinc halides.

Key words: trialkylpyrazine, pheromone, Negishi cross-coupling, pyrazine triflate, organozinc halide

Pyrazine molecules are ubiquitously distributed in plants, animals, and microorganisms, and show a wide range of biological properties.¹ Among them, a large number of trialkylpyrazines have been identified mostly as flavor components in many food stuffs² and trail pheromone components in various species of ants.³ Recent explorations have revealed that volatiles in the headspace bouquets released by marine bacteria are composed of several alkylated and methoxypyrazines as the major components.⁴ It has been suggested that the bacterial volatiles are possibly involved in cell-to-cell communication with other organisms in order to influence populations, communities, and ecosystems.⁵ Due to their interesting structural features and unique sensory properties, regiocontrolled synthesis of trialkylpyrazines has elicited much attention.

Unsymmetrical alkylpyrazines were frequently produced from pyrazines by the treatment of alkyllithium or Grignard reagents, while the outcomes were overwhelmed with the disadvantages such as incomplete conversion and formation of isomers, which may cause troublesome purification of the product.^{2a,3a,6} To overcome the regioselectivity problem, much effort has been expended on multistep synthetic sequences. Thus, the regiocontrolled synthesis of trialkylpyrazines has been conducted based upon thermal electrocyclization of 1-hydroxy-1,4-diazahexatrienes, pyrolysis of 2-norbornyl-3,5-dimethylpyrazines, zirconium-mediated reactions of alkylpyrazines and alkynes, and cyclocondensation of 2-aminoketones with 2-nitroketones using an electron-transfer reagent, respectively.⁷

Transition-metal-catalyzed cross-coupling reaction is one of the most powerful tools for constructing new carbon–

carbon bonds in the synthesis of highly functionalized aromatic and heterocyclic compounds. The cross-coupling reaction usually proceeds under mild reaction conditions and the key advantage is the exceptional tolerance of most functional groups. Thus, a number of functionalized pyrazine molecules have been prepared by transition-metal-catalyzed cross-coupling reaction using halopyrazines,^{4a,8} and also by direct arylation process using pyrazine *N*-oxides⁹ and pyrazines.¹⁰ Even though the cross-coupling reactions work very effectively for the synthesis of 2,5-dimethylpyrazine analogues,^{4a,8a,c} the synthesis of 5-alkyl-2,3-dimethylpyrazines has not been documented clearly up to date.^{3a} Herein, we wish to report nickel-catalyzed synthesis of 5-alkyl-2,3-dimethylpyrazines from the corresponding pyrazine triflate with organozinc or Grignard reagents.¹¹

Halopyrazines are very important starting materials for their ability to undergo nucleophilic substitution and transition metal-catalyzed cross-coupling reactions. Most frequently, chloropyrazines are prepared by halogenation of pyrazine *N*-oxides by treating with phosphoryl chloride under refluxing conditions.^{4a} However, 5-chloropyrazine has been obtained in very low yield from 2,3-dimethylpyrazine, since the *N*-oxide affords two regioisomeric chlorides, in which chlorination takes place preferentially on the methyl group over the ring.¹² Our attention was turned to a facile synthesis of 5-hydroxypyrazine as an alternative.^{13,14} Although the condensation 1,2-dicarbonyls with α -amino acid amides was commonly employed for the synthesis of 5-hydroxy-2,3-dimethylpyrazine (**3**), this route also suffered from low yields.¹⁴ Therefore, fine-tuning of the reaction conditions was necessary in order to improve the yield.

When glycineamide hydrochloride (**2**) was used without pretreatment with alkali, the yield was considerably lower as previously described. Surprisingly, when glycineamide was regenerated from its hydrochloride by addition of aqueous sodium hydroxide in methanol prior to mixing with butane-1,2-dione (**1**), the yield was greatly improved as 81%. A similar phenomenon was previously observed in the selective formation of 5-hydroxypyrazines.¹⁵ The reason is unclear; however, it is interesting to note that the readily available nitrogen dominate the formation of the products at very initial conditions in terms of selectivity and reactivity. Some efforts on the chlorination of **3** (e.g., POCl₃, POCl₃/PCl₅, POCl₃–pyridine, SOCl₂/benzotriazole) were unsuccessful, in turn triflation was readily

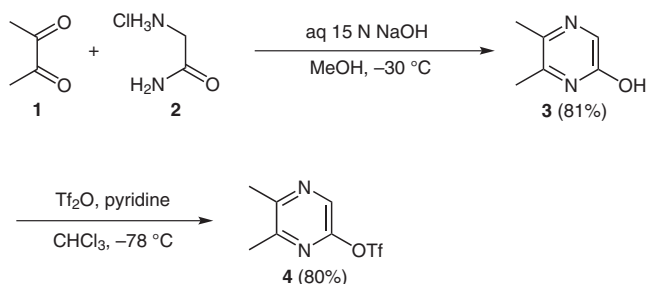
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achieved by the treatment of trifluoromethanesulfonic anhydride in the presence of pyridine as shown in Scheme 1.



Scheme 1 Synthesis of pyrazine triflate (**4**)

The triflate **4** was readily cross-coupled with different types of organometallics by the catalysis of [1,3-bis(di-phenylphosphino)propane]dichloronickel(II) [$\text{NiCl}_2(\text{dppp})$] and tetrakis(triphenylphosphine)palladium(0) [$\text{Pd}(\text{PPh}_3)_4$] to afford 5-substituted 2,3-dimethylpyrazines and the results are summarized in Tables 1 and 2. All reactions were carried out using 3 equivalents of organometallic reagents in the presence of 5 mol% of the catalyst in tetrahydrofuran where the employed reagents were all available from commercial suppliers. The Negishi cross-coupling reaction of **4** was first investigated using primary and secondary dialkylzinc (R_2Zn) with $\text{NiCl}_2(\text{dppp})$ as catalyst. The reactions with dialkylzincs proceeded efficiently leading to the formation of the desired 5-alkylpyrazines, however the yield was moderate (Table 1, entries 1–3). When alkylzinc halide (RZnX) was used instead, the result showed an increased yield of the corresponding trialkylpyrazines (entries 4–9). All synthesized trialkylpyrazines **5a–f** were volatile under reduced pressure, and thus obtained from chromatographic fractions by slow evaporation in a stream of nitrogen. Even though the use of *tert*-butylzinc bromide was unsuccessful, the primary and secondary alkylzinc halides consistently gave superior results. On the other hand, when Grignard reagents were used in place of zinc, the reactions were incomplete and the yields decreased considerably (entries 10 and 11). Thus, it was found that alkylzinc halides participated more efficiently in the nickel-catalyzed Negishi cross-coupling with pyrazine triflate.

Whereas we evaluated the same strategy using the palladium catalyst, a dramatic difference was observed in the cases of secondary alkylzinc bromides. The Negishi cross-coupling with isopropylzinc bromide under the palladium catalyst resulted in a chromatographically separable mixture of 5-isopropylpyrazine **5c** and 5-propylpyrazine **5d** in 18 and 38% yield, respectively (Table 2, entry 2). The coupling with *sec*-butylzinc also resulted in accompanying alkyl group isomerization from secondary to primary, in which 5-butylpyrazine (**5f**) was predominant (entry 3).

In turn, considering the number of observations of alkyl group isomerization in metal-catalyzed cross-coupling reactions of alkenyl and aryl halides with secondary alkyl-

Table 1 Nickel-Catalyzed Cross-Coupling Reactions of **4** with Organometallic Reagents^a

Entry	RM	Product 5	Yield (%) ^b
1	Me_2Zn	5a	38
2	Et_2Zn	5b	45
3		5c	63
4	MeZnBr	5a	61
5	EtZnBr	5b	71
6		5c	72
7	$\text{CH}_3\text{CH}_2\text{CH}_2\text{ZnBr}$	5d	74
8		5e	74
9	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{ZnBr}$	5f	81
10	EtMgBr	5b	38
11		5c	35

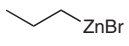
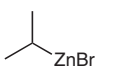
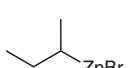
^a All reactions were performed with pyrazine triflate **4** (1 mmol), organometallic (RM, 3 mmol), $\text{NiCl}_2(\text{dppp})$ (5 mol%) in THF (3 mL).

^b Isolated yield.

metals, it is of great value to develop such site-selective cross-coupling reactions. Currently, palladium and nickel complexes are known to catalyze isomerization of secondary alkyl reagents to primary ones through a β -hydride elimination/hydrometallation sequence (path b),¹⁶ which thus compete with desired reductive elimination process (path a), as illustrated in Scheme 2. It is interesting to note that nickel-catalyzed Negishi cross-coupling reactions

with secondary alkylzinc halides bearing β -hydrogen produced the corresponding 5-alkylpyrazines as the sole products in good yields, without any isomerization of the alkyl groups (entries 6 and 8, Table 1).

Table 2 Palladium-Catalyzed Cross-Coupling Reactions of **4** with Alkylzinc Bromides^a

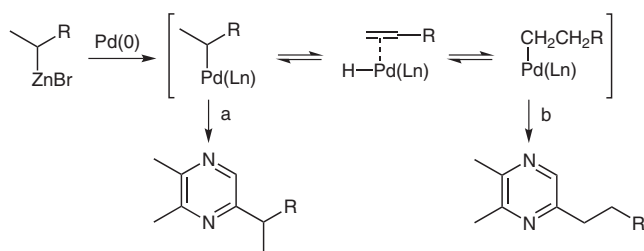
$ \begin{array}{c} \text{RZnBr (3 equiv)} \\ \text{Pd(PPh}_3)_4 \text{ (5 mol\%)} \\ \text{THF, r.t., 12 h} \end{array} \rightarrow $			
Entry	RZnX	Product 5	Yield (%) ^b
1		5d	46
2		5c + 5d	56 ^c
3		5e + 5f	43 ^d

^a All reactions were performed with pyrazine triflate **4** (1 mmol), RZnBr (3 mmol), Pd(PPh₃)₄ (5 mol%) in THF (3 mL).

^b Isolated yield.

^c Combined yield of **5c** (18%) and **5d** (38%).

^d Combined yield of **5e** (15%) and **5f** (28%).



Scheme 2 Alkyl group isomerization of secondary alkylzinc halides with Pd(PPh₃)₄

The scope of this protocol for synthesis of structurally diversified pyrazine derivatives was demonstrated next. All alkyl- and arylzinc bromides used were prepared by transmetalation of the corresponding Grignard reagents with activated zinc bromide in tetrahydrofuran,¹⁷ and were used in situ for the coupling reactions. A range of alkyl and arylzinc halides containing electronically and sterically demanding groups proved to be compatible with this operation as shown in shown in Table 3. Among them are trail pheromone components **5g–i**, eliciting trail-following behavior in various species of ants.^{3a} It was also found that cycloalkyl, aromatic, and heteroaromatic substrates were uniformly cross-coupled without any problem.

In summary, we have examined the nickel- and palladium-catalyzed cross-coupling reactions of pyrazine triflate using organozinc and Grignard reagents. It was found that 5-substituted 2,3-dimethylpyrazines was efficiently prepared by nickel-catalyzed Negishi cross-coupling reaction with alkyl and arylzinc reagents, and the alkyl group isomerization took place in palladium-catalyzed cross-coupling reactions with secondary alkylzinc halides. Thus, we have disclosed a simple procedure for the prep-

aration of trisubstituted pyrazines via organozinc mediated cross-coupling under the nickel-catalyst, and also achieved a significant improvement for the synthesis of 5-hydroxypyrazine from the condensation of glycinamide with butane-1,2-dione. This provides a ready access to regiocontrolled synthesis of trialkylpyrazines, which show interesting sensory properties in nature.

Table 3 Nickel-Catalyzed Cross-Coupling Reactions of **4** with Alkyl- and Arylzinc Bromides^a

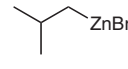
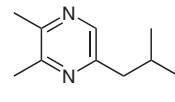
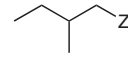
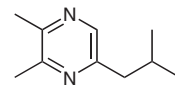
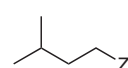
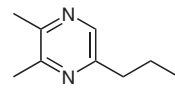
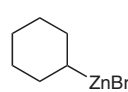
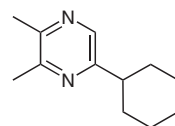
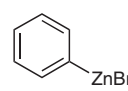
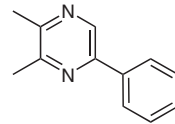
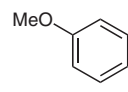
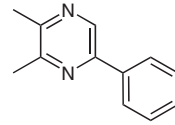
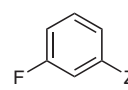
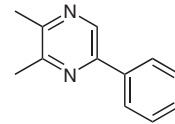
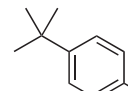
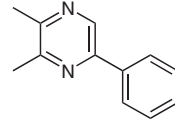
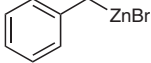
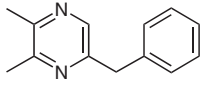
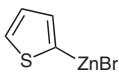
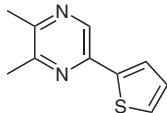
$ \begin{array}{c} \text{RZnBr (3 equiv)} \\ \text{NiCl}_2(\text{dppp}) \text{ (5 mol\%)} \\ \text{THF, r.t., 12 h} \end{array} \rightarrow $			
Entry	RZnBr ^b	Product (5)	Yield (%) ^c
1			78
2			83
3			82
4			72
5			85
6			73
7			78
8			62

Table 3 Nickel-Catalyzed Cross-Coupling Reactions of **4** with Alkyl- and Arylzinc Bromides^a (continued)

$ \begin{array}{c} \text{RZnBr (3 equiv)} \\ \text{NiCl}_2(\text{dppp}) \text{ (5 mol\%)} \\ \hline \text{4} \xrightarrow{\text{THF, r.t., 12 h}} \text{5} \end{array} $			
Entry	RZnBr ^b	Product (5)	Yield (%) ^c
9		 5o	82
10		 5p	74

^a All reactions were performed with pyrazine triflate **4** (1 mmol), RZnBr (3 mmol), NiCl₂(dppp) (5 mol%) in THF (3 mL).

^b All alkyl- and arylzinc bromides were prepared by transmetalation of the corresponding Grignard reagents (1 equiv) with anhyd ZnBr (1 equiv) in THF.

^c Isolated yield.

All of the chemicals used in this work were of analytical grade purchased from commercial suppliers and used without further purification. The reactions were monitored by TLC with Merck silica gel 60 F₂₅₄ TLC glass plates and the products were purified by flash chromatography on Merck silica gel 60 (particles size 0.040–0.063 mm) using a glass column. Melting points were measured on a Clarkson IA9200 Electrothermal Melting Point apparatus and are uncorrected. ¹H and ¹³C NMR spectra were recorded at 300 MHz and 75 MHz, respectively, on a Jeol Eclipse FT 300 MHz Spectrometer in CDCl₃ as solvent using TMS as internal standard and chemical shifts are expressed as δ ppm. Mass spectra were detected by electron impact (EI) and obtained on an Agilent 1100 Series VLL or Jeol MStation JMS 700 mass spectrometer.

5-Hydroxy-2,3-dimethylpyrazine (**3**)

To a suspension of glycine hydrochloride (**2**; 1.12 g, 10.1 mmol) in MeOH (20 mL) at –30 to –40 °C was added a solution of aq 15 N NaOH (17 mL, 25.5 mmol) and then the mixture allowed to regenerate glycine from its hydrochloride for 10 min. A solution of **1** (1.03 g, 12 mmol) in MeOH (10 mL) was added to this mixture dropwise and the reaction mixture was maintained at –30 °C for additional 0.5 h and at r.t. for 3 h. The precipitated salts were filtered off and the filtrate was dried (MgSO₄), and concentrated to give a residue. The residue was purified by column chromatography on silica gel (CH₂Cl₂–MeOH, 15:1) to give 1.02 g (81%) of **3** as a solid; mp 198–200 °C (Lit.^{14a} mp 199–200 °C; Lit.^{14b} mp 201–202 °C).

¹H NMR (300 MHz, CDCl₃): δ = 13.8 (1 H, s), 8.03 (1 H, s), 2.34 (3 H, s), 2.31 (3 H, s).

¹³C NMR (75 MHz, CDCl₃): δ = 158.4, 143.7, 133.7, 131.6, 18.6, 16.4.

EI-MS: *m/z* (%) = 124 (M⁺, 78), 111 (4), 99 (2), 95 (100), 81 (36), 69 (12), 54 (23).

2,3-Dimethyl-5-trifluoromethanesulfonyloxypyrazine (**4**)

To a solution of **3** (1.24 g, 10 mmol) in CHCl₃ (45 mL) at –78 °C was added Tf₂O (2.52 mL, 15 mmol) and pyridine (1.83 mL, 15.5 mmol) successively via a syringe under argon atmosphere. The re-

sultant reaction mixture was stirred for 2 h at –78 °C and at r.t. for 30 min. The reaction was quenched with ice water (20 mL) and the organic layer was extracted with CH₂Cl₂ (2 × 20 mL). The combined organic extracts were washed with brine (2 × 20 mL) and dried (MgSO₄). The solvents were filtered and removed by reduced pressure resulted residue. The crude residue was purified by flash silica gel chromatography (Et₂O–hexane, 15:85) to give **4** (1.82 g, 80%) as an oil.

¹H NMR (300 MHz, CDCl₃): δ = 8.28 (s, 1 H), 2.60 (s, 3 H), 2.56 (s, 3 H).

¹³C NMR (75 MHz, CDCl₃): δ = 153.6, 151.9, 150.7, 133.1, 118.5 (d, *J* = 318.6 Hz), 21.7, 21.5.

EI-MS: *m/z* (%) = 256 (M⁺, 100), 246 (5), 228 (42), 95 (11), 82 (56), 69 (47).

EI-HRMS: *m/z* calcd for C₇H₇F₃N₂O₃S: 256.0129; found: 256.0123.

Nickel- and Palladium-Catalyzed Cross-Coupling of **4** with Organometallics; General Procedure

In a flame-dried flask, the triflate **4** (1 mmol) and NiCl₂(dppp)/Pd(PPh₃)₄ (5 mol%) was dissolved in THF (3 mL) at r.t. under argon atmosphere. To this solution was added the corresponding organometallic reagent (3 equiv) and the mixture was stirred for 12 h at r.t. The reaction mixture was concentrated and to this residue 5% aq NaHCO₃ (5 mL) was added via a syringe. The precipitate was filtered off and the mixture was extracted with Et₂O (2 × 10 mL). The combined organic layers were dried (MgSO₄), and concentrated under a gentle stream of N₂ or evaporated under reduced pressure (bath temp below 20 °C). The residue was subjected to column chromatography on silica gel (hexane–Et₂O, 8:2) and the solvent was evaporated to dryness under a gentle stream of N₂ to afford the desired product as an oil (Tables 1 and 2).

2,3,5-Trimethylpyrazine (**5a**)^{6a,7a} (Table 1, entry 4)

Yield: 74.5 mg (61%); yellow oil.

¹H NMR (300 MHz, CDCl₃): δ = 8.16 (s, 1 H), 2.51 (s, 3 H), 2.50 (s, 3 H), 2.48 (s, 3 H).

¹³C NMR (75 MHz, CDCl₃): δ = 151.3, 150.4, 149.1, 140.8, 22.1, 21.6, 21.1.

EI-MS: *m/z* (%) = 122 (M⁺, 100), 111 (22), 97 (29), 69 (47), 57 (83).

5-Ethyl-2,3-dimethylpyrazine (**5b**)^{6a,7a} (Table 1, entry 5)

Yield: 96.7 mg (71%); yellow oil.

¹H NMR (300 MHz, CDCl₃): δ = 8.16 (s, 1 H), 2.75 (q, *J* = 7.4 Hz, 2 H), 2.51 (s, 3 H), 2.50 (s, 3 H), 1.29 (t, *J* = 7.7 Hz, 3 H).

¹³C NMR (75 MHz, CDCl₃): δ = 155.1, 151.4, 149.4, 139.9, 28.3, 22.1, 21.7, 14.0.

EI-MS: *m/z* (%) = 136 (M⁺, 18), 121 (20), 111 (27), 97 (39), 85 (44), 71 (63), 57 (87).

EI-HRMS: *m/z* calcd for C₈H₁₂N₂: 136.1000; found: 136.1002.

5-Isopropyl-2,3-dimethylpyrazine (**5c**) (Table 1, entry 6)

Yield: 108.1 mg (72%); yellow oil.

¹H NMR (300 MHz, CDCl₃): δ = 8.16 (s, 1 H), 3.02 (sept, *J* = 7.3 Hz, 1 H), 2.51 (s, 3 H), 2.49 (s, 3 H), 1.30 (d, *J* = 7.4 Hz, 6 H).

¹³C NMR (75 MHz, CDCl₃): δ = 158.7, 151.2, 149.4, 138.8, 33.7, 22.4, 22.2, 21.7.

EI-MS: *m/z* (%) = 150 (M⁺, 3), 149 (100), 145 (12), 140 (18), 132 (22), 123 (29).

EI-HRMS: *m/z* calcd for C₉H₁₄N₂: 150.1157; found: 150.1159.

2,3-Dimethyl-5-propylpyrazine (**5d**)^{3c} (Table 1, entry 7)

Yield: 111.2 mg (74%); yellow oil.

^1H NMR (300 MHz, CDCl_3): δ = 8.14 (s, 1 H), 2.72 (t, J = 7.4 Hz, 2 H), 2.51 (s, 3 H), 2.50 (s, 3 H), 1.74 (sext, J = 7.5 Hz, 2 H), 0.97 (t, J = 7.3 Hz, 3 H).

^{13}C NMR (75 MHz, CDCl_3): δ = 153.9, 151.5, 149.3, 140.5, 37.2, 23.2, 22.2, 21.7, 13.9.

EI-MS: m/z (%) = 150 (M^+ , 18), 121 (20), 111 (27), 97 (39), 85 (44), 71 (63), 57 (87).

EI-HRMS: m/z calcd for $\text{C}_8\text{H}_{12}\text{N}_2$: 150.1157; found: 150.1156.

5-(sec-Butyl)-2,3-dimethylpyrazine (5e) (Table 1, entry 8)

Yield: 121.5 mg (74%); yellow oil.

^1H NMR (300 MHz, CDCl_3): δ = 8.12 (s, 1 H), 2.75 (q, J = 7.2 Hz, 1 H), 2.51 (s, 3 H), 2.50 (s, 3 H), 1.80–1.57 (m, 2 H), 1.27 (d, J = 7.0 Hz, 3 H), 0.84 (t, J = 7.0 Hz, 3 H).

^{13}C NMR (75 MHz, CDCl_3): δ = 157.9, 151.4, 149.3, 139.6, 40.9, 29.7, 22.2, 21.7, 20.2, 12.1.

EI-MS: m/z (%) = 164 (M^+ , 23), 156 (4), 149 (100), 136 (62), 122 (22), 113 (6), 109 (16).

EI-HRMS: m/z calcd for $\text{C}_{10}\text{H}_{16}\text{N}_2$: 164.1313; found: 164.1312.

5-Butyl-2,3-dimethylpyrazine (5f) (Table 1, entry 9)

Yield: 133.0 mg (81%); yellow oil.

^1H NMR (300 MHz, CDCl_3): δ = 8.08 (s, 1 H), 2.66 (t, J = 7.8 Hz, 2 H), 2.45 (s, 3 H), 2.44 (s, 3 H), 1.62 (quint, J = 7.8 Hz, 2 H), 1.34 (sext, J = 7.2 Hz, 2 H), 0.87 (t, J = 7.2 Hz, 3 H).

^{13}C NMR (75 MHz, CDCl_3): δ = 154.0, 151.4, 149.2, 140.4, 34.9, 31.9, 22.5, 22.0, 21.6, 13.9.

EI-MS: m/z (%) = 164 (M^+ , 8), 149 (54), 135 (18), 122 (100), 105 (14), 97 (14).

EI-HRMS: m/z calcd for $\text{C}_8\text{H}_{12}\text{N}_2$: 164.1310; found: 164.1311.

Alkyl- and Arylzinc Bromides; General Procedure

In a flame-dried flask under argon, anhyd ZnBr_2 (0.45 g, 2 mmol) was charged and activated using hot gun under high vacuum. The flask was refilled with argon and the solid was dissolved in anhyd THF (4 mL) at 0 °C. To this cooled solution was added the respective Grignard reagent (2 mmol) in THF– Et_2O slowly and stirred for 30 min at the same temperature. After 10 min at r.t., the solution was used for the next coupling reaction.

Nickel-Catalyzed Cross-Coupling Reactions of Pyrazine Triflate with Organozinc Bromides; 2,3-Dimethyl-5-(2-methylbutyl)pyrazine (5h);^{3a} Typical Procedure

In a flame-dried flask, the triflate **4** (0.22 g, 1 mmol) and $\text{NiCl}_2(\text{dppp})$ (27 mg, 0.05 mmol) was dissolved in THF (3 mL) at r.t. under argon atmosphere. To this solution was added 2-methylbutylzinc bromide solution (0.5 M in THF, 6 mL, 3 mmol) and the mixture was stirred for 12 h at r.t. During the addition of the zinc reagent, the reaction mixture turned to black. The mixture was concentrated and to this residue was added 5% aq NaHCO_3 (5 mL) via a syringe. The precipitate was filtered off and the mixture was extracted with Et_2O (2×10 mL). The combined organic layers were dried (MgSO_4) and concentrated under a gentle stream of N_2 or evaporated under reduced pressure (bath temp below 20 °C). The residue was subjected to column chromatography on silica gel (Et_2O –hexane, 8:2) and the solvent was evaporated to dryness under a gentle stream of N_2 to afford 148.0 mg (83%) of **5h** as a yellow oil.

^1H NMR (300 MHz, CDCl_3): δ = 8.06 (s, 1 H), 2.68 (dd, J = 5.7, 13.4 Hz, 2 H), 2.46 (s, 3 H), 2.42 (s, 3 H), 1.90–1.73 (m, 1 H), 1.45–1.29 (m, 1 H), 1.24–1.09 (m, 1 H), 0.91–0.81 (m, 6 H).

^{13}C NMR (75 MHz, CDCl_3): δ = 153.2, 151.4, 149.1, 141.1, 42.2, 33.9, 29.3, 22.1, 21.6, 18.8, 11.3.

EI-MS: m/z (%) = 178 (M^+ , 64), 149 (100), 121 (28), 109 (21).

EI-HRMS: m/z calcd for $\text{C}_{11}\text{H}_{18}\text{N}_2$: 178.1470; found: 178.1475.

5-Isobutyl-2,3-dimethylpyrazine (5g)^{3a}

Yield: 128.1 mg (78%); yellow oil.

^1H NMR (300 MHz, CDCl_3): δ = 8.10 (s, 1 H), 2.58 (d, J = 7.3 Hz, 2 H), 2.51 (s, 3 H), 2.50 (s, 3 H), 2.14–1.98 (m, 1 H), 0.93 (d, J = 6.3 Hz, 6 H).

^{13}C NMR (75 MHz, CDCl_3): δ = 153.0, 151.4, 149.2, 141.0, 44.1, 28.9, 22.3, 22.0, 21.5.

EI-MS: m/z (%) = 164 (M^+ , 23), 156 (4), 149 (100), 136 (62), 122 (22), 113 (6), 109 (16).

EI-HRMS: m/z calcd for $\text{C}_{10}\text{H}_{16}\text{N}_2$: 164.1313; found: 164.1313.

5-Isopentyl-2,3-Dimethylpyrazine (5i)^{3a}

Yield: 146.2 mg (82%); yellow oil.

^1H NMR (300 MHz, CDCl_3): δ = 8.14 (s, 1 H), 2.72 (t, J = 7.3 Hz, 2 H), 2.51 (s, 3 H), 2.50 (s, 3 H), 1.68–1.52 (m, 3 H), 0.95 (d, J = 6.0 Hz, 6 H).

^{13}C NMR (75 MHz, CDCl_3): δ = 154.3, 151.4, 149.2, 140.4, 38.9, 33.2, 28.0, 22.5, 22.1, 21.6.

EI-MS: m/z (%) = 178 (M^+ , 6), 149 (41), 122 (100), 97 (36), 83 (40), 69 (55), 57 (55).

EI-HRMS: m/z calcd for $\text{C}_{11}\text{H}_{18}\text{N}_2$: 178.1470; found: 178.1479.

5-Cyclohexyl-2,3-dimethylpyrazine (5j)

Yield: 137.0 mg (72%); yellow oil.

^1H NMR (300 MHz, CDCl_3): δ = 8.15 (s, 1 H), 2.73–2.59 (m, 1 H), 2.50 (s, 3 H), 2.49 (s, 3 H), 1.98–1.70 (m, 4 H), 1.62–1.20 (m, 6 H).

^{13}C NMR (75 MHz, CDCl_3): δ = 157.9, 151.2, 149.4, 139.1, 43.9, 32.7, 26.5, 26.0, 22.1, 22.0.

EI-MS: m/z (%) = 190 (M^+ , 40), 161 (15), 149 (19), 135 (100), 122 (66).

EI-HRMS: m/z calcd for $\text{C}_{12}\text{H}_{18}\text{N}_2$: 190.1470; found: 190.1467.

2,3-Dimethyl-5-phenylpyrazine (5k)

Yield: 156.6 mg (85%); white solid; mp 58–59 °C.

^1H NMR (300 MHz, CDCl_3): δ = 8.71 (s, 1 H), 7.97 (d, J = 7.1 Hz, 2 H), 7.52–7.39 (m, 3 H), 2.61 (s, 3 H), 2.57 (s, 3 H).

^{13}C NMR (75 MHz, CDCl_3): δ = 151.8, 150.6, 149.5, 138.4, 136.9, 129.3, 129.0, 126.8, 22.4, 21.9.

EI-MS: m/z (%) = 184 (M^+ , 89), 169 (8), 143 (5), 114 (5), 102 (100), 76 (10).

EI-HRMS: m/z calcd for $\text{C}_{12}\text{H}_{12}\text{N}_2$: 184.1000; found: 184.1005.

5-(4-Methoxyphenyl)-2,3-dimethylpyrazine (5l)

Yield: 156.4 mg (73%); white solid; mp 63–64 °C.

^1H NMR (300 MHz, CDCl_3): δ = 8.65 (s, 1 H), 7.93 (d, J = 8.2 Hz, 2 H), 7.01 (d, J = 8.9 Hz, 2 H), 3.86 (s, 3 H), 2.58 (s, 3 H), 2.55 (s, 3 H).

^{13}C NMR (75 MHz, CDCl_3): δ = 160.8, 151.7, 149.7, 149.3, 137.8, 129.6, 128.1, 114.5, 55.5, 22.4, 21.8.

EI-MS: m/z (%) = 214 (M^+ , 100), 199 (27), 171 (12), 132 (31), 102 (6), 89 (13).

EI-HRMS: m/z calcd for $\text{C}_{13}\text{H}_{14}\text{N}_2\text{O}$: 214.1101; found: 214.1101.

5-(3-Fluorophenyl)-2,3-dimethylpyrazine (5m)

Yield: 157.8 mg (78%); white solid; mp 67–68 °C.

^1H NMR (300 MHz, CDCl_3): δ = 8.67 (s, 1 H), 7.74–7.69 (m, 2 H), 7.45–7.38 (m, 1 H), 7.12–7.06 (m, 1 H), 2.37 (s, 3 H), 2.35 (s, 3 H).

^{13}C NMR (75 MHz, CDCl_3): δ = 163.4 (d, J = 253.1 Hz), 151.9, 151.2, 148.2, 139.3, 138.3, 130.6, 122.2, 116.3, 113.9, 22.4, 21.9.

EI-MS: m/z (%) = 202 (M^+ , 100), 159 (7), 119 (48).

EI-HRMS: m/z calcd for $\text{C}_{12}\text{H}_{11}\text{FN}_2$: 202.0906; found: 202.0906.

5-(4-*tert*-Butylphenyl)-2,3-dimethylpyrazine (5n)

Yield: 149.0 mg (62%); yellow oil.

¹H NMR (300 MHz, CDCl₃): δ = 8.67 (s, 1 H), 7.90 (d, *J* = 8.3 Hz, 2 H), 7.50 (d, *J* = 8.3 Hz, 2 H), 2.59 (s, 3 H), 2.56 (s, 3 H), 1.35 (s, 9 H).¹³C NMR (75 MHz, CDCl₃): δ = 152.5, 150.1, 143.0, 138.2, 134.1, 126.5, 125.9, 125.3, 34.8, 31.4, 22.4, 21.8.EI-MS: *m/z* (%) = 240 (M⁺, 34), 225 (100), 197 (11), 115 (14), 102 (6), 66 (4).EI-HRMS: *m/z* calcd for C₁₆H₂₀N₂: 240.1626; found: 240.1628.**5-Benzyl-2,3-dimethylpyrazine (5o)**

Yield: 162.6 mg (82%); yellow oil.

¹H NMR (300 MHz, CDCl₃): δ = 8.12 (s, 1 H), 7.33–7.14 (m, 5 H), 4.09 (s, 2 H), 2.51 (s, 3 H), 2.48 (s, 3 H).¹³C NMR (75 MHz, CDCl₃): δ = 152.6, 149.8, 140.9, 138.9, 129.1, 128.8, 128.4, 126.7, 41.7, 22.2, 21.7.EI-MS: *m/z* (%) = 198 (M⁺, 61), 197 (100), 115 (42), 104 (13), 91 (33).EI-HRMS: *m/z* calcd for C₁₆H₂₀N₂: 198.1157; found: 198.1159.**2,3-Dimethyl-5-(2-thienyl)pyrazine (5p)**

Yield: 140.8 mg (74%); white solid; mp 57–58 °C.

¹H NMR (300 MHz, CDCl₃): δ = 8.64 (s, 1 H), 7.61 (d, *J* = 1.2 Hz, 1 H), 7.41 (d, *J* = 2.3 Hz, 1 H), 7.14–7.10 (dd, *J* = 0.9, 3.0 Hz, 1 H), 2.56 (s, 3 H), 2.54 (s, 3 H).¹³C NMR (75 MHz, CDCl₃): δ = 151.8, 150.2, 145.3, 141.9, 136.7, 128.2, 127.8, 124.8, 22.1, 21.8.EI-MS: *m/z* (%) = 190 (M⁺, 100), 147 (7), 107 (74), 68 (6), 57 (4).EI-HRMS: *m/z* calcd for C₁₀H₁₀N₂S: 190.0565; found: 190.0561.**Acknowledgment**

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